

# SWAPNANIL BALA

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## EDUCATION

### Northeastern University, Khoury College of Computer Sciences

M.S. in Data Science | Seeking Spring 2027 Co-op

Boston, MA

Sep 2025 – Dec 2027 (Expected)

### Maulana Abul Kalam Azad University of Technology

B.Tech in Electronics and Communication Engineering

West Bengal, India

Aug 2019 – Aug 2023

## TECHNICAL SKILLS

**Languages** TypeScript, Python, JavaScript, SQL

**AI & LLM Eng.** OpenAI GPT-4o Vision, Anthropic Claude API, MediaPipe, multimodal inference, prompt design, structured response parsing

**Machine Learning** PyTorch, Scikit-Learn, XGBoost, transformer fine-tuning (DistilBERT, RoBERTa), NLP, SHAP explainability

**Data & Analytics** Pandas, NumPy, EDA, hypothesis testing (t-test, chi-square, ANOVA), Matplotlib, Seaborn

**Frontend** React, Next.js, PWAs (service workers, web workers), i18n, Lighthouse performance optimization

**Backend & APIs** Node.js, REST API design, edge functions, OAuth 2.0, Postgres Row-Level Security, rate limiting, LRU/TTL caching

**Data & DevOps** PostgreSQL (NeonDB, Supabase), Zod schema validation, Vitest/Jest, Git, Docker, Vercel

## PROJECTS

### Lagna Atelier — Local-First Vedic Astrology Engine

Web (PWA) | Deployed

TypeScript · Next.js · astronomy-engine · Zod · GPT-4o Vision · MediaPipe · Vercel

- Built a ~63K-line local-first PWA with no backend server, computing every chart client-side and persisting user data in localStorage — zero server round-trips for the core product.
- Hand-wrote a sidereal ephemeris layer on astronomy-engine implementing IAU-2006 precession, supporting 6 ayanamsha models and 6 house systems.
- Engineered 15+ derived calculation engines covering 23 divisional charts, Vimshottari dasha periods, and 51 yoga detections.
- Designed a Zod-validated rules-as-data DSL for yoga definitions, with a Monte Carlo harness measuring each rule's empirical rarity across simulated chart populations.
- Shipped a multimodal palm-reading feature on GPT-4o Vision with MediaPipe hand landmarking, owning image ingestion, prompt design, and structured response parsing behind rate-limited edge functions.
- Optimized delivery with service workers, web workers for heavy computation, and LRU/TTL caching; localized to 6 languages; backed by 561 tests.

### Robust Health — AI Fitness & Wellness Platform

Full-Stack Web | Deployed

TypeScript · Next.js · Anthropic API · NeonDB (Postgres) · Supabase Auth · Vercel

- Shipped a multi-role web application with member and trainer dashboards, tiered subscriptions, and CSRF protection.
- Served personalized, AI-generated workout plans through the Anthropic API via Next.js API routes backed by NeonDB.
- Implemented authentication with Supabase Row-Level Security and OAuth 2.0 providers (Google, GitHub, Discord).
- Cut Lighthouse-measured load time ~50% on web and ~67% on mobile through next/image compression and route-level code splitting.

### Fake News Classifier — Transformer Fine-Tuning & Explainability

Applied NLP Research

Python · PyTorch · DistilBERT · RoBERTa · SHAP · Scikit-Learn

- Fine-tuned DistilBERT and RoBERTa on the LIAR dataset for multi-class misinformation detection.
- Applied SHAP for token-level explainability, surfacing which phrases drove each classification.
- Benchmarked both transformers against a TF-IDF and logistic regression baseline, delivering a structured precision, recall, and F1 comparison across all three approaches.

## EXPERIENCE

### P2G Mobility Tech Private Limited (Pointo) — EV-Financing Startup

Kolkata, India

Program Analyst Intern, Technology

Nov 2024 – Jan 2025

- Cleaned and consolidated operational Excel datasets feeding internal dashboards, resolving missing values, recovering lost source files, and standardizing records for weekly reporting.
- Wrote SQL against the production database and built recurring operational reports for the Project Lead.
- Supported field customer acquisition, pitching the pay-to-own financing program to e-rickshaw operators and framing the lithium-ion upgrade against legacy lead-acid batteries.

## CERTIFICATIONS

**Google Advanced Data Analytics Professional Certificate** — TikTok claims classification: NLP pipeline flagging policy-violating content at ~99.5% recall (5 misclassifications across 3,817 test cases). Waze churn prediction: EDA and hypothesis testing (t-test, chi-square, ANOVA); tuned logistic regression, Random Forest, and XGBoost with GridSearchCV.